

The Spectral Predictor for Grokking

What it measures, why it was tried, and what result it gave in our benchmark.

Project: *A Unified Benchmark of Grokking Predictors in Neural Networks* (M.Sc. AI thesis).

Predictor 3 of 9.

How to use this document. This document is written so that it can be given to any AI assistant for question-and-answer. Every formula is followed by a line "In words", so that the meaning is clear even if the formula is not read properly. The last section has a glossary and a list of common questions with answers.

1. The one-paragraph summary

The Spectral predictor takes a trained model at one moment of training. It looks at the hidden representation of the model on the training data. It then asks one question: **how many directions of this representation are needed to explain the labels?** If only a few directions are enough, the model is well aligned with the task. The idea is that this alignment should improve just before grokking, so it could warn us in advance. In our experiment this did **not** happen. The signal changed only slightly, and its best value came at or after grokking. So the Spectral predictor is recorded as a **closed negative result**.

2. Background you need first

2.1 What is grokking?

Grokking means this. The model first memorises the training data. Training accuracy becomes almost 100%, but test accuracy stays low. After a long time, test accuracy suddenly jumps to almost 100%. This sudden late jump is called grokking.

2.2 What is a "grokking predictor"?

A grokking predictor is a measurable quantity, computed from the model while it trains. It should show a clear change **before** the test-accuracy jump. Then it can predict that grokking is coming. Our thesis compares nine such predictors under one common protocol. The order is: L2 Norm, Dropout, Spectral, AGE, HTSR Alpha, Correlation Traps, Weight-PCA, Higher-MI, Commutator Defect.

2.3 Our experimental setup (the "substrate")

Item	Value
Task	Modular addition: predict $(a + b) \bmod 113$
Model	One-layer transformer, 4 attention heads, $d_{\text{model}} = 128$, small initialisation (std 0.0707)
Optimiser	AdamW, learning rate 10^{-3} , betas (0.9, 0.98), weight decay 1.0
Training data	30% of all pairs: $N = 3830 = 0.3 \times 113^2$. Full-batch training.
Length	40,000 epochs, 5 random seeds
Checkpoints	24 checkpoints, spaced log-uniformly over training
Grok epoch	The first epoch at which test accuracy is above 0.9

The Spectral predictor is a **checkpoint-only** predictor. This means it is computed from a saved copy of the model at one epoch. It does not need the training run itself.

3. Where the idea comes from

The Spectral predictor follows Canatar, Bordelon and Pehlevan (2021), "*Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks*", Nature Communications 12:2914.

Their main idea, in simple terms: a model learns fast those parts of the target function that lie along the **top eigen-directions** of its kernel. A model generalises well when the target function has most of its "power" in these top directions. They call this **task-model alignment**.

We borrowed this idea and asked: during grokking, does the hidden representation of the network become better aligned with the labels, and does this happen before the test accuracy jumps?

4. What exactly is measured (step by step)

Take one frozen checkpoint. Follow these steps.

Step 1: Collect the representation

For each of the $N = 3830$ **training** examples, run the model forward. Record the final hidden vector at the "=" position (token index 2). This vector has 128 numbers. It is captured by a forward hook on the layer output_head. Stack all vectors as rows:

$$\Phi \in \mathbb{R}^{N \times 128}$$

In words: Φ is a table with one row per training example and 128 columns. Each row is what the model "thinks" about that example, just before it gives the answer.

Then centre the columns, that is, subtract the mean of each column. The centred matrix is Φ_c .

Step 2: Build the kernel (Gram) matrix

$$K = \Phi_c \Phi_c^T \in \mathbb{R}^{N \times N}$$

In words: K_{ij} is the dot product of the centred representations of example i and example j . A large value means the model treats the two examples as similar.

Step 3: Eigen-decomposition of the kernel

$$K = U \text{diag}(\eta_1, \eta_2, \dots, \eta_N) U^T, \quad \eta_1 \geq \eta_2 \geq \dots \geq 0$$

In words: η_k are the eigenvalues, sorted from largest to smallest. The columns of U are the eigenvectors. A large η_k means the representation varies a lot along that direction. (This is Equation 3 in Canatar et al.)

Step 4: Project the labels on the eigenvectors

Let $Y \in \mathbb{R}^{N \times 113}$ be the one-hot labels (113 classes). Centre it to get Y_c . Then compute:

$$W = U^T Y_c, \quad w_k^2 = \|W_k\|^2, \quad p_k = \frac{w_k^2}{\sum_j w_j^2}$$

In words: w_k^2 is the "task power" along the k -th eigen-direction. It says how much of the label information lies along that direction. p_k is the same quantity written as a fraction, so that all p_k add up to 1. (Equations 4 and 5 in Canatar et al.)

Step 5: Cumulative power

$$C(k) = \sum_{j \leq k} p_j$$

In words: $C(k)$ is the fraction of label power that is contained in the top k eigen-directions. This curve is called the task-model alignment curve. (Equation 6 in Canatar et al.)

5. The numbers we record from this curve

Signal	Definition	Meaning
k_{90}	Smallest k such that $C(k) \geq 0.90$	How many directions are needed to explain 90% of the labels. Small means good alignment.
k_{95}	Smallest k such that $C(k) \geq 0.95$	Same, for 95%.
alignment_score	$\sum_k \tilde{\eta}_k p_k$, where $\tilde{\eta}_k$ are the normalised eigenvalues	Large when label power sits on the large-eigenvalue directions. Large means good alignment.
entropy	$-\sum_k p_k \log p_k$	How widely the label power is spread over directions. Small means concentrated.

In words: for k_{90} , lower is better alignment. For alignment_score, higher is better alignment. For entropy, lower means the label power is concentrated in few directions.

6. The hypothesis being tested

Before grokking, the model has only memorised. Its representation is not related to the true rule. So the label power is spread over almost all 3830 directions, and k_{90} is large.

If the model starts learning the real rule, the label power should move to the top eigen-directions. Then:

- k_{90} should **fall**.
- alignment_score should **rise**.
- This should happen **at or before** the grok epoch.

The test compares $k_{90_min_epoch}$ (the epoch where k_{90} is lowest) and $alignment_max_epoch$ (the epoch where alignment_score is highest) against $grok_epoch$. If they come clearly before the grok epoch, the predictor works.

7. What we found

Verdict: CLOSED NEGATIVE. The Spectral predictor does not give an early warning of grokking in this benchmark.

- k_{90} fell only by about 2 to 3%. It went from about 3448 to about 3350–3400, out of 3830 possible directions. This is a very small change.
- Entropy went from about 8.24 to about 7.94.
- alignment_score rose from about 2×10^{-4} to about 8.5×10^{-3} . This rise is real. But its maximum came at or after the grok epoch in **all 5 seeds**.

Per-seed timing (epochs)

Seed	Grok epoch	k90_min_epoch	alignment_max_epoch
1	14474	25232	25232
2	6988	10040	15917
3	10418	25232	39999
4	10193	39999	39999
5	24021	39999	39999

Mean grok epoch = 13218.8, standard deviation = 5900.6. In every seed, both signals reach their best value **after** the grok epoch. A predictor must come before the event. So this signal is late, not early.

Note: the seed numbering 1 to 5 above only follows the order of the recorded lists. It is used here for reading convenience.

8. Why did it fail?

Reason 1: wrong regime. Canatar's theory is for the **lazy (kernel, NTK) regime**. In this regime, the network features stay almost fixed, and only the readout is learned. Our network is small, with small initialisation and large weight decay. It is in the **feature-learning regime**.

Reason 2: what grokking really is here. On modular addition, the network builds a Fourier-multiplication circuit. This is a structured change in a few frequency components. It is not a global collapse of the representation to a few directions. Because of this, the label power does not concentrate in the top eigen-directions before the jump.

Reason 3: the small change is expected. A tiny drop in k_{90} is what a correct implementation should give. It does not mean there is a bug. It also does not contradict Canatar et al., because their theory was never claimed for this regime.

A short history: an earlier version of this predictor used weight SVD (stable rank and similar quantities). It was deleted on 6 September, because it was not faithful to Canatar et al. The present version follows the paper's kernel and label-projection method.

9. Implementation notes

- The Gram matrix and eigen-decomposition are computed on **CPU in float64**. Reason: Apple Silicon MPS has no eigh and no float64. (MPS also lacks the SVD operation.)
- The predictor is a **checkpoint-only plugin** in `run_nanda_benchmark.py`. It is registered in `CHECKPOINT_PREDICTOR_FUNCS` and `PREDICTOR_SUMMARY_KEY`.
- Main functions: `compute_spectral_metrics_for_checkpoint()` in `src/predictors/spectral.py`, and `save_spectral_data()` in

`src/unified_measurements.py`.

- The shared model file `transformer.py` is not owned by this predictor. It is used by training, all predictors and evaluation.

10. Where Spectral stands in the whole benchmark

Predictor	Result
L2 Norm	Closed negative. Moving-average crossover at epoch about 103-121, which is far too early.
Dropout-Variance (Salah & Yevick)	Closed negative. Peak comes after grokking.
Spectral (this document)	Closed negative. Best value at or after grokking.
AGE (Papayan, NC1)	Closed negative. Collapse comes after grokking.
HTSR Alpha	Next predictor to be worked on.

The supervisor, Prof. Rashid, confirmed on 2026-09-17 that the baseline set is L2, Dropout, Spectral and AGE, and that it is complete. The predictor code for the earlier predictors is still in the project. The work has only moved to the next predictor.

11. Glossary

Term	Meaning
Grokking	Late, sudden jump in test accuracy after the training accuracy is already near 100%.
Representation	The hidden vector computed by the network for an input. Here, the 128-number vector at the "=" position.
Kernel / Gram matrix	A matrix of pairwise dot products between representations of examples.
Eigenvalue / eigenvector	For a symmetric matrix K : a direction u and a number η with $Ku = \eta u$. The eigenvalue tells how strong the matrix is in that direction.
Spectrum	The list of eigenvalues, sorted from large to small.
Spectral bias	The tendency of a model to learn the top eigen-directions of its kernel first.
Task power	w_k^2 : the part of the label signal that lies along eigen-direction k .
Task-model alignment	How much of the label power lies in the top eigen-directions. Measured by $C(k)$.
Lazy / kernel / NTK regime	Network features stay almost fixed during training. The network behaves like a fixed kernel method.
Feature learning regime	The hidden features change strongly during training. Grokking on modular addition belongs here.
Checkpoint	A saved copy of the model at a given epoch.
Closed negative	The predictor was fully tested, and it did not predict grokking early.

12. Questions and answers

Q1. In one line, what does the Spectral predictor measure?

It measures how many top eigen-directions of the model's representation kernel are needed to explain the label function. This is summarised by k_{90} , alignment_score and entropy.

Q2. Which data is used: training or test?

Only the 3830 **training** examples are used. The labels used are the true training labels, in one-hot form.

Q3. Which layer's output is used?

The final hidden state at the "=" position (index 2), just before the output head. It has 128 dimensions.

Q4. Why are the labels centred?

Centring removes the constant (mean) part, so that only the variation of the labels is analysed. The same is done for the representation.

Q5. What does a small k_{90} mean?

It means that 90% of the label power is contained in a few top directions. This is good task-model alignment.

Q6. Did k_{90} become small during our experiment?

No. It changed only from about 3448 to about 3350–3400 out of 3830. This is a change of only about 2–3%.

Q7. The alignment_score did rise by a large factor. Why is this not a success?

A predictor must reach its signal **before** the grok epoch. The maximum of alignment_score came at or after the grok epoch in all 5 seeds. So it is a late indicator, not an early warning.

Q8. Does this failure mean that Canatar et al. is wrong?

No. Their theory is for the lazy (kernel) regime. Grokking here is in the feature-learning regime. So the theory is simply not expected to apply directly.

Q9. Does this failure mean the code has a bug?

There is no evidence of a bug. A small change in k_{90} is what a correct implementation is expected to give in this setting.

Q10. Why is the computation done on CPU?

The Gram matrix is 3830 by 3830. Its eigen-decomposition needs eigh and float64. Apple Silicon MPS supports neither, so the step runs on CPU.

Q11. How is the grok epoch defined?

It is the first epoch at which test accuracy is greater than 0.9.

Q12. What happens next in the project?

The next predictor is HTSR Alpha (Predictor 5). The Spectral code stays in the project. It is simply not being worked on now.

13. Reference

A. Canatar, B. Bordelon, C. Pehlevan. *Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks*. Nature Communications 12, 2914 (2021).

Project file for detailed notes: Spectral_Predictor_Task_Model_Alignment.md (in the literature notes folder).